

Eren Homburg

Curriculum Vitae



Personal information

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Date of Birth 12.09.2002

Education

- Sep 2022 - Aug 2025 **B.Sc Computer Science,**
ETH Zürich, Zürich, Switzerland
- Thesis: 'DBox Plus: An AI-Assisted Code Learning Tutor' supervised by Prof. Dr. April Wang
 - Core Courses: Compiler Design, Human-Computer Interaction, Visual Computing, Introduction to Machine Learning, and Rigorous Software Engineering
- Sep 2017- Jun 2021 **High School,**
Liceo Cantonale di Mendrisio, Mendrisio, Switzerland
Physics and applied Mathematics

Projects

- Feb 2025 – Jul 2025 **DBox Plus – An AI-Assisted Code Learning Tutor,**
Bachelor Thesis supervised by Prof. April Wang, (ETH Zürich Switzerland),
dbox.com 
- **Skills:** React, TypeScript, Python, User-Centered Design, Git, ~~La~~TeX, Figma, Docker
 - Built the Decomposition Box (DBox) from scratch using React, including a complete frontend and a redesigned user interface
 - Designed and implemented system extensions to teach abstraction by identifying and generalizing recurring programming patterns
 - Integrated large language models to support learners through interactive step trees that foster co-decomposition and independent thinking
 - Deployed backend on Cloudflare Workers using R2 and KV storage, with rate limiting, file streaming, and resource caching
 - Applied a user-centered design process, conducted a usability study (N=7), and evaluated the system using SUS and likeliness scores; performed quantitative and qualitative analysis, including visualization of user feedback and behavioral patterns

Nov 2024 – Dec 2024

Urban AI – Scalable Web Application for Urban Insight,
Fundamentals of Web Engineering, (ETH Zürich Switzerland),
[github.com/repo](#) 

- **Skills:** React, JavaScript, TypeScript, Cloudflare, OpenLayers, HTML, CSS, Google Maps API, Git, $\LaTeX$
- Built a **full-stack web application** to generate architectural and civil engineering insights from user-uploaded images
- Designed an interactive map (OpenLayers) to visualize geo-curated datasets from 6 major cities, including dynamic coloring and filtering
- Developed a responsive frontend (React) with image upload portal, token system, user authentication, and Stripe integration
- Deployed backend on Cloudflare Workers using R2 and KV storage, with rate limiting, file streaming, and resource caching
- Integrated Google Maps Street View API, Grounding DINO, and SAM for street-level insight extraction and segmentation
- Ensured security and scalability via CORS handling, Cloudflare edge caching, and secure email routing via contact form

Oct 2024 – Dec 2024

Family FairShare: Balance the Invisible Work,
Human Computer Interaction, (ETH Zürich Switzerland),
[github.com/repo](#) 

- **Skills:** Figma, Miro, PHP, Python, User Research, A/B Testing, Git, $\LaTeX$
- Designed and prototyped a mobile app to visualize and fairly redistribute household mental load using user-centered design methods
- Ideated with 6-3-5 Brainwriting and affinity diagrams; benchmarked apps like Trello for visualization inspiration
- Developed three prototype iterations (low-, mid-, and high-fidelity) using Figma; incorporated feedback from usability tests and A/B testing
- Engineered dynamic task visualization tools (e.g., heatmaps, radar charts, load timelines) to aid in workload awareness and planning
- Evaluated gamification's impact via user studies; removed features that increased mental load or reduced fairness in real-world group tests

Feb 2024 – Jul 2024

Flight Jet Simulator, VR, Universal Render Pipeline (URP),
Visualization, Simulation and Interaction - Virtual Reality I, (ETH Zürich Switzerland)

- **Skills:** Blender, Unity, C#
- Designed a custom jet model in Blender and integrated it into Unity
- Developed immersive game mechanics and physics interactions for VR

Experience

Nov 2025 - Present

GenAI R&D Engineer (Student Assistant),

ETH Zürich – GenAI-R&D for Future-Proof Teaching, Zürich, Switzerland

- Developing and improving the front-end of the open-source *EthelApp* platform (desktop & mobile) using **React** and **TypeScript**
- Implementing user interfaces and interaction flows that support 20,000+ students across STEM courses at ETH
- Collaborating with researchers and instructors to design and iterate on educational AI tools
- Ensuring smooth integration between front-end components and **Python**-based back-end systems
- Maintaining clear documentation and contributing to the open-source community under **GNU-GPL**

Skills

Programming
Languages &
Frameworks

Python (Pandas, NumPy, PyTorch), Java, C++, C#, C, Assembly, Haskell, OCaml, JavaScript/TypeScript (React, Node.js, Chart.js), PHP, HTML/CSS, MySQL

Tools & Platforms	Git, Docker (basic), $\LaTeX$, Cloudflare, Blender, Unity, Miro, Figma, Canva, Microsoft Office
Hardware & Embedded	FPGAs (basic), Assembly
UX & HCI	User Research, Usability Testing (SUS, A/B Testing), Survey Design, Wireframing, Affinity Mapping, Task Flows, Statistical Analysis (Python)
Languages	German (Native), Italian (Native), English (Professional Proficiency), Turkish (Professional Proficiency), French (Limited Professional Proficiency)